

HYEONGHUN KIM

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EDUCATION

University of California San Diego

Ph.D. in Mechanical Engineering, GPA: 3.93/4.00

La Jolla, CA

Fall 2022–Present

Handong University

B.S. in Mechanical Engineering, GPA: 4.03/4.50 (Magna cum laude)

Minor in Electrical and Control Engineering

Pohang, South Korea

March 2015–February 2021

RESEARCH INTEREST

Reduced-order modeling, physics-based modeling, scientific machine learning, nonlinear dynamics, stochastic system and modeling

PUBLICATIONS

1. **Kim, H** and Kramer, B. (2025). Physically consistent predictive reduced-order modeling by enhancing Operator Inference with state constraints, *Journal of Computational Physics*, 543, 114418. DOI: 10.1016/j.jcp.2025.114418.
2. Kang, S, **Kim, H**, and Kramer, B. (2026). Parametric Operator Inference to Simulate the Purging Process in Semiconductor Manufacturing, *IEEE Transactions on Semiconductor Manufacturing*, Early Access. DOI: 10.1109/TSM.2026.3694257.

GRADUATE RESEARCH PROJECTS

Kramer Research Group

Graduate Research Assistant

Advisor: Dr. Boris Kramer

La Jolla, CA

September 2023–Present

- Strong Turbulence and Rogue Wave Modeling using Machine-Learning Assisted Predictions (Ongoing)
 - Developing data-driven stochastic reduced-order models of experimentally measured capillary wave turbulence using stochastic Operator Inference. We aim to learn low-dimensional stochastic differential equations (ROMs) directly from noisy experimental data and capture key turbulent features of capillary wave dynamics.
- Parametric Operator Inference to simulate the purging process in plasma-enhanced chemical vapor deposition (PECVD)
 - Using a scientific machine learning method, Operator Inference, we developed a parametric surrogate model that predicts the purging flow field within the PECVD chamber. This, trained on 36% of the data and tested on 64%, demonstrated accuracy across the entire parameter domain, with a maximum error of 9.32%, providing an effective particle contamination control solution in semiconductor manufacturing.
- Physically predictive reduced-order modeling by enhancing Operator Inference with state constraints
 - We improved the standard Operator Inference by augmenting state constraints inherent in the full-order char combustion model. This showed superior accuracy and stability to other stability-enhancing approaches, such as eigenvalue reassignment and constrained optimization.

It extrapolated 200% beyond the training regime while maintaining physical consistency across the entire spatiotemporal domain and achieving a 3,584-fold speedup in online computation.

INTERNSHIP

ASML

Graduate Intern

Computational Fluid Dynamics Team

San Diego, CA

July 2025–September 2025

- The first project involved further developing a steady-state reduced-order modeling algorithm that uses machine learning techniques (linear regression, k-nearest neighbors, Kriging, and Gaussian Process regression) for the CFD simulation data, to determine the optimal method that minimizes prediction errors. The automation algorithm was integrated and packaged via MATLAB App.
- The second project focused on developing a transient reduced-order modeling algorithm that uses highly transient full flow-field of EUV source CFD data. By applying parametric Dynamic Mode Decomposition (DMD), we achieved predictions of crucial flow quantities, including localized species concentrations relevant to reducing tin deposition on the EUV source machine.

UNDERGRADUATE RESEARCH PROJECTS

Multiagent Information Fusion and Control Lab

Undergraduate Research Assistant

Advisor: Dr. Won-Sang Ra

Pohang, South Korea

July 2020–December 2020

- Fast feature extraction using CNN and Hough Transformed lidar point cloud data
 - Enhanced the existing feature extraction method that used Hough transform, by (a) training the ideal features from HT using a convolutional neural network (b) using Chebyshev sampling to reduce the amount training data with respect to the range of look angles.

Energy Engineering Lab

Undergraduate Research Assistant

Advisor: Dr. Kwon-Yeong Lee

Pohang, South Korea

December 2019–June 2020

- Experimental study and development of a heat-resistance prediction program for automotive lamp
 - Collaborated with *SL Co.*, developed a MATLAB algorithm utilizing heat transfer and multivariate regression to predict heat resistance and inner surface temperature of automotive lamps from initial hardware conditions (CFD data provided by the company).

RELEVANT GRADUATE COURSEWORK

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|--|----------------------------|
| • Numerical Linear Algebra and ODE simulation | • Linear Control Design |
| • Numerical Methods for Differential Equations | • Nonlinear Systems |
| • Computational Fluid Dynamics | • Fluid Mechanics 1 |
| • Uncertainty Quantification | • Fluid Mechanics 2 |
| • Linear Systems Theory | • Conduction Heat Transfer |
| • Optimal Estimation | • Mass Transfer |

COMPUTER SKILLS

- Coding: MATLAB, Python, LaTeX, Bash, Git
- Software: ANSYS, MFiX

TEACHING EXPERIENCE

Graduate Teaching Assistant at UCSD
Signals and Systems (MAE 143A)

Winter 2025

Undergraduate Tutor at Handong
Calculus 2

Fall 2020

MILITARY SERVICE

Republic of Korea Army (Army Aviation Operations Command) Icheon, South Korea
Sergeant February 2016–November 2017

Military specialization in strategy and information.
Led and managed a 13-member squad for four months.
Led a mentoring program at Icheon Danwol Elementary School, teaching math and English to underrepresented students.

HONORS AND AWARDS

Magna Cum Laude	Feb. 2021
Academic Excellence Scholarship in School of Mechanical and Control Engineering	2020, 2019
National Science and Engineering Scholarship (Four semesters full tuition)	Spring 2019
Korea Student Aid Foundation	Spring 2018, Fall 2015